

BUTTS AAF, CO, USAWMO: **724680**Lat: **38.678N**Lon: **104.757W**Elev: **1779**StdP: **81.70**Time Zone: **-7.00 (NAM)**Period: **07-19**WBAN: **94015****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-16.7	-13.8	-22.8	0.6	-3.5	-20.2	0.8	-2.2	16.0	3.5	14.0	0.7	2.3	330	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.1	33.8	15.0	32.4	15.0	30.9	15.0	18.1	26.4	17.6	26.2	16.9	25.6	5.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	14.1	19.9	15.1	13.4	19.4	14.3	12.7	19.5	58.4	26.5	56.2	26.2	54.0	25.5	20.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.5	11.3	9.3	-22.1	36.0	3.4	1.0	-24.5	36.7	-26.5	37.3	-28.4	37.9	-30.9	38.7		
(5)			-22.4	19.7	3.3	0.6	-24.8	20.1	-26.8	20.5	-28.6	20.8	-31.1	21.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	-0.1	0.6	5.7	8.8	13.5	20.3	22.6	21.4	18.0	10.6	5.0	-0.6
	DBStd	9.55	5.64	6.19	5.44	4.92	4.60	3.43	2.54	2.50	3.63	5.33	5.47	6.12
	HDD10.0	1382	313	266	151	79	22	0	0	0	2	57	161	331
	HDD18.3	3206	571	497	391	286	159	19	3	6	48	239	399	588
	CDD10.0	1581	1	2	19	44	129	309	392	352	242	77	12	1
	CDD18.3	363	0	0	0	0	8	79	136	99	39	1	0	0
	CDH23.3	4698	0	1	8	22	203	1158	1526	1098	608	73	1	0
	CDH26.7	1877	0	0	0	1	47	505	701	428	189	6	0	0
Wind	WSAvg	4.0	3.5	4.1	4.3	4.9	4.4	4.3	3.7	3.5	3.8	3.9	3.8	3.6
Precipitation	PrecAvg	446	12	11	25	50	59	46	82	78	35	22	14	10
	PrecMax	652	31	37	64	184	210	129	159	139	122	67	62	35
	PrecMin	206	2	1	3	3	17	6	21	10	1	1	1	3
	PrecStd	109	7	8	16	37	44	33	37	33	27	17	13	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.4	20.6	24.5	25.9	30.4	36.0	35.1	34.2	32.8	27.7	22.9	18.1
		MCWB	6.4	6.9	8.5	9.5	11.9	14.2	15.3	14.9	15.1	10.8	8.1	5.9
	2%	DB	15.1	17.0	21.6	23.6	27.9	33.6	33.8	32.4	30.4	25.5	20.4	14.8
		MCWB	4.4	5.3	7.2	8.9	11.0	13.8	15.5	15.4	13.7	10.3	7.2	4.4
	5%	DB	12.5	14.6	19.1	21.6	25.7	31.8	32.6	30.9	29.0	23.4	17.8	12.4
		MCWB	3.3	4.3	6.1	8.1	11.0	13.6	15.6	15.6	13.5	9.8	6.4	3.3
	10%	DB	9.7	11.6	16.6	19.4	23.2	29.8	31.0	29.4	27.3	20.8	15.1	9.5
		MCWB	1.9	2.9	5.4	7.4	10.7	13.6	15.9	15.8	13.1	9.0	5.0	2.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.4	7.3	8.9	11.5	14.6	18.3	19.0	19.0	17.1	13.5	8.7	6.5
		MCDB	16.9	19.5	22.7	19.7	23.7	26.9	26.1	26.9	26.4	20.9	20.6	17.1
	2%	WB	4.7	5.7	7.8	10.2	13.3	17.1	18.2	18.1	15.8	12.2	7.7	4.8
		MCDB	14.1	16.1	20.1	19.3	22.4	26.4	26.1	26.5	24.9	20.4	19.4	14.0
	5%	WB	3.5	4.5	6.7	9.1	12.5	16.1	17.7	17.5	15.1	10.8	6.6	3.5
		MCDB	12.1	14.2	17.8	18.7	21.3	25.5	26.3	25.8	24.5	20.1	17.1	11.8
	10%	WB	2.0	3.0	5.6	8.1	11.6	15.3	17.0	16.8	14.3	9.8	5.4	2.1
		MCDB	9.0	11.3	15.6	17.9	20.4	24.9	25.9	25.0	23.5	19.6	14.5	9.4
Mean Daily Temperature Range	5% DB	MDBR	14.2	14.2	15.4	14.5	14.2	16.0	15.1	14.5	15.1	14.8	14.9	13.6
		MCDBR	18.0	18.7	19.0	18.4	18.5	19.0	17.7	17.1	17.7	18.8	18.6	18.2
		MCWBR	9.8	9.7	8.9	7.9	7.1	6.4	5.4	5.4	6.3	7.9	8.9	9.9
	5% WB	MCDBR	17.3	18.4	18.1	16.7	15.6	15.6	14.4	14.0	15.3	16.1	17.5	17.4
		MCWBR	9.6	9.8	8.7	7.5	6.6	6.4	5.2	5.3	5.9	7.4	8.7	9.8
Clear-Sky Solar Irradiance	taub		0.235	0.239	0.261	0.287	0.306	0.320	0.350	0.336	0.300	0.266	0.244	0.237
	taud		2.624	2.642	2.576	2.512	2.518	2.495	2.466	2.495	2.569	2.640	2.626	2.613
	Ebn at Noon		979	1015	1011	993	971	953	921	930	953	966	958	951
	Edh at Noon		67	76	90	103	104	107	109	104	90	75	66	63
All-Sky Solar Radiation	RadAvg		2.76	3.71	4.95	5.98	6.61	7.34	6.89	6.29	5.45	4.11	3.02	2.43
	RadStd		0.14	0.24	0.36	0.36	0.49	0.34	0.29	0.22	0.28	0.24	0.17	0.13

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+70	+23

Nomenclature: See separate page